

TEST 1

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BTECH CSE 'B'

ROLLNO : 20

Q1. Counter — Counter.jsx

```
import React, { useState } from "react";

function Counter() {

  const [count, setCount] = useState(0);

  return (

    <div>

      <h2>Counter: {count}</h2>

      <button onClick={() => setCount(count + 1)}>

        Increment

      </button>

      <button

        onClick={() => setCount(count - 1)}

        disabled={count === 0}

      >

        Decrement

      </button>

      <button onClick={() => setCount(0)}>

        Reset

      </button>

    </div>

  );
}
```

```
export default Counter;
```

Q2. Todo List — `TodoList.jsx`

```
import React, { useState } from "react";

function TodoList() {

  const [task, setTask] = useState("");

  const [todos, setTodos] = useState([]);

  const addTodo = () => {

    if (task.trim() === "") return;

    setTodos([...todos, task]);

    setTask("");

  };

  const deleteTodo = (index) => {

    setTodos(todos.filter((_, i) => i !== index));

  };

  return (

    <div>

      <h2>Todo List</h2>

      <input

        type="text"

        value={task}

        onChange={(e) => setTask(e.target.value)}

        placeholder="Enter a task"

      />

      <button onClick={addTodo}>Add</button>

      <ul>

        {todos.map((todo, index) => (

          <li key={index}>
```

```

      {todo}

      <button onClick={() => deleteTodo(index)}>

        Delete

      </button>

    </li>

  )))

</ul>

</div>

);

}

export default TodoList;

```

Q3. Login Toggle — LoginToggle.jsx

```

import React, { useState } from "react";

function LoginToggle() {

  const [loggedIn, setLoggedIn] = useState(false);

  return (

    <div>

      {!loggedIn ? (

        <button onClick={() => setLoggedIn(true)}>

          Login

        </button>

      ) : (

        <div>

          <h2>Welcome!</h2>

          <button onClick={() => setLoggedIn(false)}>

            Logout

          </button>

        </div>

```

```
        </div>

    )}

</div>

);

}

export default LoginToggle;
```

Q4. Live Search Filter — LiveSearch.jsx

```
import React, { useState } from "react";

function LiveSearch() {

    const [search, setSearch] = useState("");

    const names = [

        "Rahul",

        "Aman",

        "Priya",

        "Neha",

        "Riya",

        "Arjun",

        "Karan",

        "Ananya",

        "Rohit",

        "Sneha"

    ];

    const filteredNames = names.filter((name) =>

        name.toLowerCase().includes(search.toLowerCase())

    );

    return (

        <div>
```

```

<h2>Live Search</h2>

<input
  type="text"
  value={search}
  onChange={(e) => setSearch(e.target.value)}
  placeholder="Search name"
/>

<ul>
  {filteredNames.length > 0 ? (
    filteredNames.map((name, index) => (
      <li key={index}>{name}</li>
    ))
  ) : (
    <p>No results found</p>
  )}
</ul>
</div>

);
}

export default LiveSearch;

```

Q5. Stopwatch — Stopwatch.jsx

```

import React, { useEffect, useState } from "react";

function Stopwatch() {

  const [seconds, setSeconds] = useState(0);

  const [running, setRunning] = useState(false);

  useEffect(() => {

    let interval;

```

```

if (running) {

  interval = setInterval(() => {

    setSeconds((prevSeconds) => prevSeconds + 1);

  }, 1000);

}

return () => {

  clearInterval(interval);

};

}, [running]);

const reset = () => {

  setRunning(false);

  setSeconds(0);

};

return (

  <div>

    <h2>Stopwatch: {seconds} seconds</h2>

    <button onClick={() => setRunning(true)}>

      Start

    </button>

    <button onClick={() => setRunning(false)}>

      Pause

    </button>

    <button onClick={reset}>

      Reset

    </button>

  </div>

);

}

```

```
export default Stopwatch;
```

App.jsx to run all 5 components

```
import React from "react";

import Counter from "./Counter";

import TodoList from "./TodoList";

import LoginToggle from "./LoginToggle";

import LiveSearch from "./LiveSearch";

import Stopwatch from "./Stopwatch";

function App() {

  return (

    <div>

      <h1>Full Stack Web Development Practical Test</h1>

      <hr />

      <Counter />

      <hr />

      <TodoList />

      <hr />

      <LoginToggle />

      <hr />

      <LiveSearch />

      <hr />

      <Stopwatch />

    </div>

  );

}

export default App;
```